

MAKILESH M

AI/ML Engineer · Data Scientist · LLMs & Voice AI Specialist

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PROFESSIONAL SUMMARY

Oracle-certified AI/ML Engineer with hands-on experience building LLM applications, RAG pipelines, multi-agent systems, and real-time Voice AI solutions. Proven ability to develop production-oriented AI systems with strong focus on low-latency inference, scalable architectures, and practical deployment across cloud and local environments.

SKILLS

AI & Machine Learning: LLMs, RAG, Multi-Agent Systems, NLP, Transformers, Generative AI, Deep Learning, Model Fine-tuning, MLOps, Prompt Engineering, Vector Embeddings

Frameworks & Libraries: LangChain, LangGraph, CrewAI, AutoGen, PyTorch, TensorFlow, Scikit-learn, Hugging Face, spaCy, OpenCV, Streamlit, Whisper, Twilio

Cloud & Tools: AWS (EC2, S3, Lambda), Azure OpenAI, Docker, FastAPI, Flask, Git, SQL, MySQL, Pinecone, FAISS, ChromaDB, Weaviate, Milvus, n8n

Languages: Python, C++, C, Java, JavaScript, HTML, CSS, React.js, RESTful APIs

WORK EXPERIENCE

AI Developer Intern | Shamla Tech Solutions (Sep 2025 - Feb 2026)

- Engineered and deployed an internal **multi-platform lead scraping pipeline** (Reddit, Discord, Slack, LinkedIn) with a 3-stage heuristic pre-filter (spam detection, job post classification, provider filtering) achieving **94.6%** lower LLM API calls, with GPT-4-turbo with Ollama fallback and batch processing across concurrent async scrapers.
- Developed an **AI Voice Calling Agent** (Twilio + Pinecone) enabling fully automated outbound calls with zero manual intervention. Implemented persistent conversation memory via Pinecone vector storage, delivering context-aware dynamic responses based on prior interaction history and validated through response quality and latency benchmarking
- Built a full-duplex **Cross-Platform Meeting Assistant** with real-time 10-speaker diarization and streaming transcription across Zoom, Google Meet, Microsoft Teams, and in-person sessions, featuring wake-word activation for on-demand contextual AI expert responses powered by RAG.
- Technologies Used:** Python, LangChain, Twilio API, RealtimeSTT/TTS, Whisper, Pinecone, pyannote.audio, OpenAI / Gemini APIs, Selenium, Playwright, PRAW, Discord.py, Slack SDK, AsyncIO, Flask

PROJECTS

Agentic RAG System (Jan 2026 - Feb 2026) - github.com/Makilesh/agentic_rag_system

- Engineered a multi-agent document Q&A platform using a 7-agent LangGraph state machine with a self-corrective retrieval loop that automatically rewrites low-quality queries (score < 0.7) and re-retrieves up to 2 times, improving answer accuracy across multi-format documents (PDF, DOCX, PPTX, Excel, TXT)
- Implemented true hybrid search over Milvus 2.4 vector database combining dense semantic vectors (384-dim Sentence Transformers, HNSW index) with sparse BM25-style keyword vectors via Reciprocal Rank Fusion (RRF) at ~200ms retrieval latency. Deployed via Docker Compose with Gemini 2.5 Flash primary LLM and Ollama Qwen 2.5 fallback
- Technologies Used:** Python, LangGraph, LangChain, Milvus, Sentence Transformers, FastAPI, Docker, PyMuPDF

AI-Powered Voice Agent (Sep 2025 - Nov 2025) - github.com/Makilesh/voice_engine_MVP

- Engineered a production-grade real-time full-duplex voice agent achieving 200–500ms end-to-end latency (STT + LLM + TTS) with <150ms barge-in detection via RMS energy monitoring and selective echo suppression, sentiment-aware adaptive tone modulation, and a producer-consumer async architecture with Silero VAD and Whisper transcription
- Built a hybrid TTS pipeline using Kokoro-82M (40–90ms first-byte latency) with a LLM fallback chain, dynamic audio queue sizing, and memory-aware concurrency management to prevent OOM failures under sustained load.
- Technologies Used:** Python, Asyncio, Threading, Whisper, RealtimeSTT, Kokoro, PyAudio, OpenAI, psutil, httpx

CERTIFICATIONS & ACHIEVEMENTS

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional - 2025
- Oracle Cloud Infrastructure 2025 Certified Data Science Professional - 2025
- Udemy Complete Machine Learning, NLP Bootcamp MLOPS & Deployment - 2024
- NPTEL Python for Data Science - 2024
- IMPACT X - Finalist, Top 3 of 200+ teams, Team Lead & AI Engineer, AI Crime Detection System using YOLOv8 (2024)
- THEERKATHON - Winner, Team Lead, Won product incubation opportunity with Forge (2024)
- THIRAN - Finalist, Top 5 of 500+ teams, Team Lead & AI Engineer, AI Legal Engine for Indian Criminal Law (2025)

EDUCATION

Sri Krishna College of Engineering and Technology, Coimbatore, India

B.E. in Computer Science Engineering | CGPA: 8.0/10 | May 2026